DataExchange Data QC & Reconciliation Procedure

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Signatures

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Document Control	
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1 Introduction

1.1 Purpose

This procedure establishes the guidelines and methods for data quality checks and data reconciliation in DataExchange.

1.2 Scope

Technology and Product Management, DevOps & Software Development Teams

- Employees supporting database, ETL processes related to DataExchange.
    - ETL Developers & Database Administrators
- Employees supporting the software development and testing of DataExchange
    - Software Developers & QA Analysts

Software Implementation, Technology Consulting Team

- Employees supporting the implementation of DataExchange to clients
    - Business System Analysts & Solution Architects

1.3 Objectives

The objective of Dynamic Risk is to ensure data is verified and accurate during the ETL process via IRAS DataExchange. It is essential that we reconcile data on a regular basis to safeguard the accuracy of the data and to build confidence among users. Data reconciliation is the process of verifying and validating data collected from multiple sources to the destination database and to ensure that data moves from system-to-system without any data loss or data duplication.

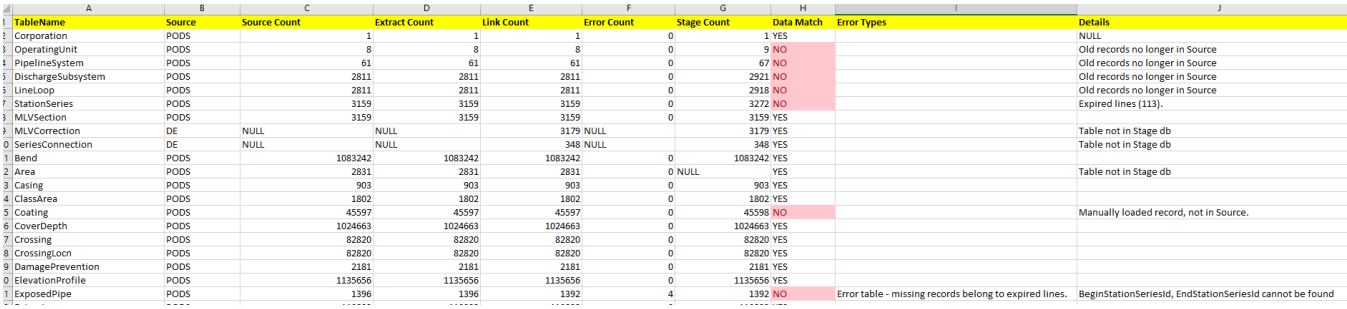
The following are some of the problems that need to be resolved:

- Missing or mismatching records
- Missing or mismatching values
- Duplicate data
- Wrong values
- Ill-formatted records
- Ill-formatted values
- Lack of sync across tables and system

2 Data QC Spreadsheet

The Data QC Spreadsheet can be accessed here: DataExchange DataQC Template.xlsx

The ETL team uses the following spreadsheet for data QC:



TableName	Source	Source Count	Extract Count	Link Count	Error Count	Stage Count	Data Match	Error Types	Details
Corporation	PODS	1	1	1	0	1	YES		NULL
OperatingUnit	PODS	8	8	8	0	9	NO		Old records no longer in Source
PipelineSystem	PODS	61	61	61	0	67	NO		Old records no longer in Source
DischargeSubsystem	PODS	2811	2811	2811	0	2921	NO		Old records no longer in Source
LineLoop	PODS	2811	2811	2811	0	2918	NO		Old records no longer in Source
StationSeries	PODS	3159	3159	3159	0	3272	NO		Expired lines (113).
MLVSection	PODS	3159	3159	3159	0	3159	YES		
MLVCorrection	DE	NULL	NULL	3179	NULL	3179	YES		Table not in Stage db
SeriesConnection	DE	NULL	NULL	348	NULL	348	YES		Table not in Stage db
Bend	PODS	1083242	1083242	1083242	0	1083242	YES		
Area	PODS	2831	2831	2831	0	NULL	YES		Table not in Stage db
Casing	PODS	903	903	903	0	903	YES		
ClassArea	PODS	1802	1802	1802	0	1802	YES		
Coating	PODS	45597	45597	45597	0	45598	NO		Manually loaded record, not in Source.
CoverDepth	PODS	1024663	1024663	1024663	0	1024663	YES		
Crossing	PODS	82820	82820	82820	0	82820	YES		
CrossingLocn	PODS	82820	82820	82820	0	82820	YES		
DamagePrevention	PODS	2181	2181	2181	0	2181	YES		
ElevationProfile	PODS	1135656	1135656	1135656	0	1135656	YES		
ExposedPipe	PODS	1396	1396	1392	4	1392	NO	Error table - missing records belong to expired lines.	BeginStationSeriesId, EndStationSeriesId cannot be found

The definitions for each column in the spreadsheet:

TableName: The IRAS table/virtual table mapped in DataExchange

Source: The source system where data is being retrieved

Source Count: Record count retrieved by running the pull query on the source system.

The following are retrieved via the Data Reconciliation script:

Extract Count: Record count for Extract tables

Link Count: Record count for the tables in the Link database after Transform process.

Error Count: Record count for error tables

Stage Count: Record count for tables in the IRAS database after Load process

Data Match: YES if record count matches, NO if data is mismatched

Error Types: Types of errors provided in error table

Details: Additional comments or details

Once you've worked through the error reports and/or discussed them with the BSA. This is when we do count comparison between the pull queries on the Source database, after Extract/Transform has been run in Link database and once Load is completed in the Stage database. If the counts aren't matching up between Source and Link, it's important to check the Err_Tables to find the discrepancy.

Note: The Data QC spreadsheet applies for DataExchange versions 6.3.2 and greater.

For DataExchange versions prior to 6.3.2 - If the counts between Stage and Link are different and Err_Tables are empty, check the Del_Load table and it's possible that SourceTracking may need to be reset and rerun DE.

The older template of the QC spreadsheet can be found here: QATemplate.xlsx

3 Data Reconciliation Script Introduction

The Data Reconciliation script can be accessed here: Data reconciliation script.sql

The script pulls the data from Extract tables, Link tables, Error tables, and Stage counts and checks if data matches. The output of the script are three tables:

1. Centerline tables (incl. Area, Bend, DistrictDivision and StateProvince if mapped)
2. Event and virtual tables

3. Tables with errors and the error types

The output of this script can be copied into the Data QC spreadsheet and provided to the BSA for further investigation if needed.

All table counts are on the assumption of a complete DataExchange Extract, Transform and Load phase. Records are deleted from the Extract tables for duplicate SourceId or invalid composite keys.

Note: The Data Reconciliation Script only works for DataExchange versions 6.3.2 and greater.

4 Source Counts and Data Checks

Data checks on the source database are important to guarantee that data loss or data duplication hasn't occurred via DataExchange.

The following will go over the methods for validating the source

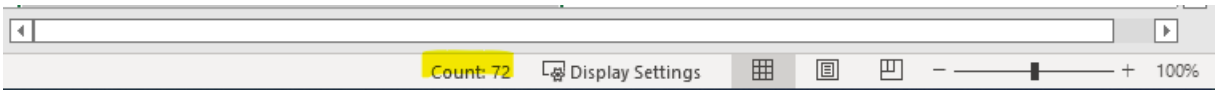
4.1 Database Source

To verify source counts from database sources (this includes Link database sources), retrieve the pull query from DataExchange and run the pull query on the source database. Take the record count and input into the Data QC spreadsheet. Ideally if the data in the source does not include duplicates, incorrect data etc. then this number should match what we have in the IRAS database.

However, data isn't always ideal so we will go over the error tables later in this procedure.

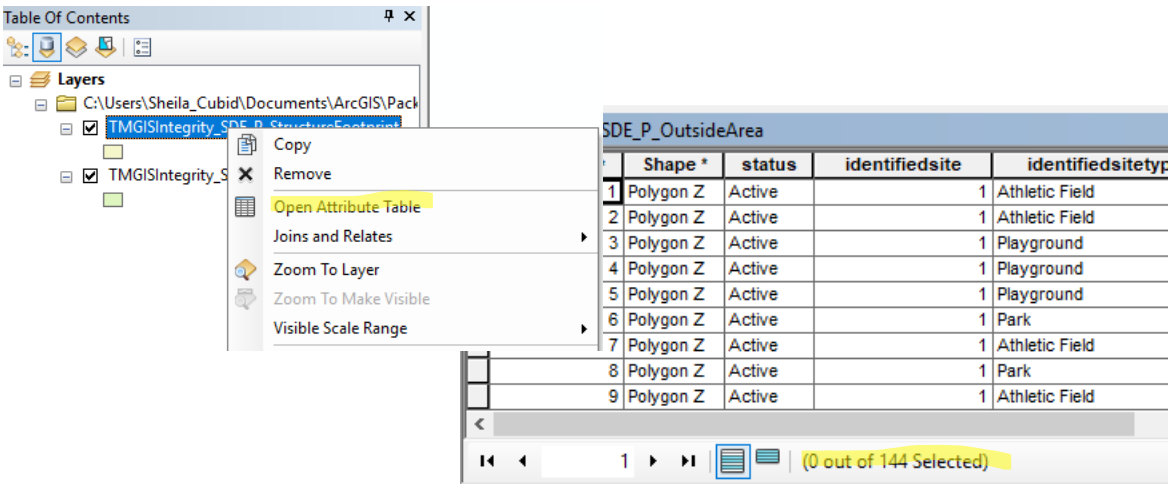
4.2 Excel Source

To verify source counts from Excel sources, open up the Excel source and take the record count by highlighting the column in the Excel sheet and Excel provides the count in the bottom right side. Input this number into the Data QC spreadsheet.



4.3 MPK Source

To retrieve counts from an MPK data source open the MPK in ESRI ArcMap, right-click on the feature class being used in the DataExchange mapping and click on Open Attribute Table. The Attribute table will show the counts in the table at the bottom of the scroll bar. Copy this number into the Data QC spreadsheet.



Note: All MPK files should have the geographic coordinate system of WGS_1984, and it should be Runtime Enabled.

5 Error Table Data Checks

DataExchange populates the error tables via the Centerprise templates. The following will go over the types of errors that can be found in the error tables and how to resolve them:

5.1 Duplicate SourceId

Error message: Duplicate SourceId

Duplicate SourceId errors occur when the value being used for SourceId is not unique in the data set.

This can be resolved by replacing the SourceId with a unique value in the data set and/or by creating a composite key. An alternative resolution would be by using the row_number() SQL function although this is not ideal since it's not created in the source database.

5.2 CompositeKeyCheck - Duplicate Record Found

Error message: Duplicate Record Found

Some child tables have a CompositeKeyCheck box which will output Duplicate Record Found in the error table.

The CompositeKeyCheck box will check the composite key is not being violated in the error table. If the CompositeKeyCheck fails, may need to verify if the values being used for the composite keys are distinct and if not, it may be an issue with the source data.

5.3 BeginStationSeriesId/EndStationSeriesId

Error message: BeginStationSeriesId/EndStationSeries cannot be null and cannot exceed 50 characters.

This error message may occur if the value being used for BeginStationSeriesId/EndStationSeriesId is not a valid Id in the StationSeries table. This can be validated by checking if the BeginStationSeriesId/EndStationSeriesId exists in the StationSeries table, StationSeries error table and source by checking StationSeries pull query.

5.4 ParentId Null

Example error message: LineLoopId cannot be null and cannot exceed 50 characters.

This error message may occur if the value being used for the parent Id is not a valid Id in the parent table. This can be validated by checking if the Id exists in the parent table, parent error table and source by checking pull query for the parent table and child table to validate the parent-child relationship.

5.5 SourceId Null

Error message: SourceId cannot be null and cannot exceed 50 characters.

This error message may occur if the value being used for the SourceId contains null values or values that exceed 50 characters. The value being used for SourceId may need to be changed in order to resolve this error either by truncating the string to be no more than 50 characters or to ensure the value being used does not contain null values and that they're all unique.

5.6 Common errors and variations

The following errors are common among all the tables and some errors occur with variations on the error message but can be resolved similarly for the error type.

5.6.1 Number has minimum and maximum value

Example: BeginStationNum has a minimum value of -99999999.99999 and a maximum value of 99999999.99999.

Example: BendLatitude has a minimum value of -99.999999999 and a maximum value of 99.999999999.

Number value being used exceeds the range for the IRAS data model of that field. This can be validated by checking if the data is being converted which could cause an issue or if the correct value is being used in the pull query.

5.6.2 String/Ind null and length

Example: BendName is longer then 30 characters. Will be truncated to 30 characters.

This error can occur if the value being used string length is longer than the characters allowed in the destination column. This can be resolved by ensuring the value is being truncated to the string length of the column in the pull query.

5.6.3 Valid date

Example: CouponReadingDate needs to be a date, with a year greater than 1754, or blank.

This can occur if the value being used for the date isn't in the correct format, is null or has an invalid year.

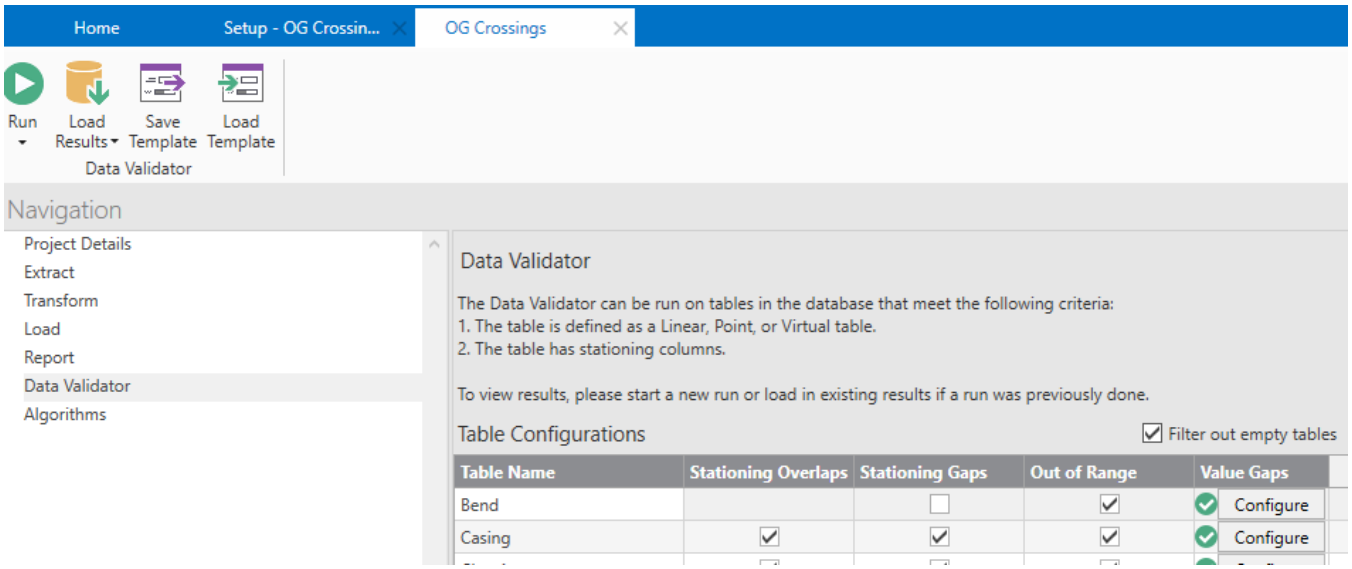
5.6.4 Valid lookup

Example: BendType value was not found in the associated lookup.

This type of error can occur if the value being used is not found in the list domain lookup.

6 Data Validator

Data Validator (previously known as Table Checker) validates the stationing data of point, linear and virtual tables where stationing applies. Data Validator is configured by the BSAs in DataExchange.



Invalid Records: Contains records where stationing is invalid in relation to the hierarchy.

Stationing Overlaps: Contains records where stationing values are duplicated or overlap on a line.

Stationing Gaps: Contains records where stationing is missing or null on a line.

7 Wholistic System Data Quality Check

The following steps are for a wholistic system data quality investigation. This isn't necessary for all data quality checks but is to be used for instances where more intensive data investigation is needed.

1. Record count compare between stage, link and source for all tables (see topics 2-4 of this procedure)

2. Mileage of the line

Mileage of the line can be calculated with the following script. Run on the Link db.

```
-- Calculate the mileage of the entire system
SELECT ou.CorporationId, c.CorporationName, sum(abs(ss.EndStationNum - ss.BeginStationNum))/1000
AS "Lenth(km)" FROM StationSeries ss
INNER JOIN LineLoop ll ON ss.LineLoopId = ll.Id
INNER JOIN PipelineSystem ps ON ps.Id = ll.PipelineSystemId
INNER JOIN OperatingUnit ou ON ou.Id = ps.OperatingUnitId
INNER JOIN Corporation c ON c.Id = ou.CorporationId
WHERE ss.EffectiveEndDate IS NULL
Group by ou.CorporationId, c.CorporationName;

SELECT sum(PipeLength) AS "Length(km)" FROM (
SELECT abs((ps.BeginStationNum * mlvBeg.MultiplierNum + mlvBeg.FactorNum) - (ps.EndStationNum *
mlvEnd.MultiplierNum + mlvEnd.FactorNum))/1000 AS PipeLength
FROM dbo.PipeSegment ps
```

```
INNER JOIN dbo.StationSeries ssBeg ON ps.BeginStationSeriesId = ssBeg.Id
INNER JOIN dbo.MLVCorrection mlvBeg ON ssBeg.Id = mlvBeg.StationSeriesId
INNER JOIN dbo.StationSeries ssEnd ON ps.EndStationSeriesId = ssEnd.Id
INNER JOIN dbo.MLVCorrection mlvEnd ON ssEnd.Id = mlvEnd.StationSeriesId

WHERE ssBeg.EffectiveEndDate IS NULL AND ssEnd.EffectiveEndDate IS NULL) tmp
```

3. Review DataExchange error report (see topic 5 of this procedure)

4. Review Table Checker report (see topic 6 of this procedure)

5. Run algorithm and review the report. Check with Config Architect

6. Check GUID lookup for centerline tables.

7. Spot check all tables and try to manually reverse engineering the data from stage and compare to the source. Make sure all conversion and lookup are correct. (We can put this into DE later on)

8. Make sure all standard DE stored procedures and functions do not have the custom process. If there is any, move them out.

8 Document Approvals

9 Revision History